

Device-Free Occupant Activity Sensing Using WiFi-Enabled IoT Devices for Smart Homes

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Abstract—Intelligent occupancy sensing is becoming a vital underpinning for various emerging applications in smart homes, such as security surveillance and human behavior analysis. However, prevailing approaches mainly rely on video camera, ambient sensors, or wearable devices, which either requires arduous deployment or arouses privacy concerns. In this paper, we present a novel real-time, device-free, and privacy-preserving WiFi-enabled Internet of Things platform for occupancy sensing, which can promote a myriad of emerging applications. It is designed to achieve an optimal tradeoff between performance and scalability. Our system empowers commercial off-the-shelf WiFi routers to collect channel state information (CSI) measurements and provides an efficient cloud server for computing via a lightweight communication protocol. To demonstrate the usefulness of our platform, an occupancy detection system is developed by exploiting the CSI curve of human presence. Furthermore, we also design an innovative activity recognition system based on our platform and machine learning techniques with high availability and extensibility. In the evaluation, the experimental results show that our platform enables these applications efficiently, with the accuracy of 96.8% and 90.6% in terms of occupancy detection and recognition, respectively.

Index Terms—Channel state information (CSI), human activity recognition, occupancy sensing, WiFi.

I. INTRODUCTION

OCCUPANCY information plays an essential role in smart building and homes in terms of energy consumption estimation, human behavior analysis, and indoor security monitoring. With the rapid development of the Internet of Things (IoT), this information can be obtained by the cognitive computing technologies, which are introduced into the

smart homes by the integration of sensors, actuators, controllers, and artificial intelligence. Various applications are enabled, of which the occupancy sensing is a significant one [1]. It refers to estimating the occupancy information and simulating the situation of indoor environments by means of the analysis of data from kinds of sensor networks deployed in smart buildings.

With occupancy sensing using IoT devices, the systems in smart building or homes are able to actively obtain the information of occupancy, such as identification, activity, and even gestures. For instance, occupancy detection and human identification help to detect the intruders automatically. In smart homes, the system can adjust the light and temperature according to the occupancy, which conserves the energy. For children and the elderly, it can monitor their activities such as falling to prevent potential hazard. Generally speaking, IoT-based occupancy sensing provides an active detected input for the smart system, so that it can make judgment or feedback accordingly.

The conventional approaches to perceive the occupants mainly resort to kinds of sensors, such as ultrasound, CO₂, and thermal sensors [2]–[4]. These works focus on coarse sensing including occupancy detection or counting [5], but as the machine learning techniques fortify the development and applications of artificial intelligence, more fine-grained sensing comes in existence. Researchers utilize sensors or wearable devices to achieve remarkable results [6]. However, the challenge in fine-grained sensing lies in finding a tradeoff between performance and availability. For instance, it is crucial to leverage a device-free manner due to convenience concern. In general, most prevailing solutions rely on dedicated sensors, such as cameras [7], RF tags [8]–[11], and accelerometers [12]–[14]. These solutions demand either toilsome installations or use of wearable devices. The camera-based methods provide highest granularity of activity recognition via computer vision algorithms, but the camera-based approaches are highly impacted by illumination and occlusion. More importantly, due to privacy concerns, it is usually not so acceptable in smart homes. Another good solution applies accelerometer equipped in most smart phones. Nevertheless, people do not necessarily carry cellphones or particular devices all the time under indoor circumstances. Radio frequency-based approaches show their advantages in low consumption and device-free manner. However, this kind of system requires many RF links in a single room, and the installation cost decreases its flexibility and availability. Some other solutions such as location-based methods need auxiliary information so

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that they cannot recognize precise activity happening in the same place.

Recently, it is observed that the motions of occupants impact the WiFi signals to some extent. Based on this insight, researchers manage to use received signal strength indicator (RSSI) [15] and channel state information (CSI) [16] to depict the human activity [17], [18]. WiFi-based technology has three advantages compared with other methods. First, it is tailored for practical deployment as a ubiquitous system. As WiFi technique has been developed rapidly, the indoor coverage of wireless networks has become more and more widespread. Thus, we do not need to install specific sensors or RF links. Second, WiFi-based approaches employ the pattern extracting of wireless signals in the physical layer, and thus do not raise issues of privacy. Third, it is able to provide relatively accurate sensing with minimal infrastructure requirements, which is a sensible tradeoff between availability and performance.

When WiFi-based methods are applied to practical use, there still remain several challenges such as scalability of platforms, lack of resolutions of wireless signals, etc. In this paper, we address these challenging problems and present a novel technology that utilizes commercial off-the-shelf (COTS) WiFi enabled IoT devices for occupancy sensing in smart homes. Our system employs the fine-grained CSI data from physical layer in the wireless connection between the transmitter (TX) and receiver (RX). It is designed for occupancy sensing with high scalability and performance, and it is composed of three parts, including an innovative IoT platform for sensing, cloud server for computing, and user end for notification. We first upgrade OpenWrt system and enable it to report exquisite CSI on the IoT devices. Then according to different application, the data can be processed in either a coarse or fine-grained way. To demonstrate the performance of our system, we then develop and realize two challenging applications, referring to occupancy detection and human activity recognition. For applications of coarse sensing such as occupancy detection, we utilize CSI curve similarity to achieve the goal. It is completely implemented on the IoT devices. For applications of fine-grained sensing such as activity recognition, we extract the features of activity by class estimated basis space singular value decomposition (CSVD) and then conduct the classification using non-negative matrix factorization (NMF) [19]. The evaluation in typical indoor environments shows that two applications achieve sensible performance in real time.

Contributions: We summarize the main contributions of this paper as follows.

- 1) We design a WiFi-enabled IoT system for occupancy sensing. To the best of our knowledge, it is the first platform which completely integrates CSI-based WiFi devices, cloud server, and user end. It makes a good tradeoff between performance and availability.
- 2) We upgrade the OpenWrt operating system with CSI reporting for COTS routers. The fine-grained CSI data and scalable platforms provides a good opportunity to promote existing applications of WiFi-based systems.
- 3) To demonstrate our framework, we elaborate a coarse sensing case, occupancy detection, based on similarity

measurement of CSI curves, and realize it on the IoT device. As far as we know, it is the first WiFi-based application purely implemented on the router.

- 4) For fine-grained sensing, we present an innovative CSVD-NMF-based activity recognition algorithm. It provides a solution to automatically extract features and conduct classification in an efficient manner.
- 5) We evaluate both the detection and recognition systems in terms of accuracy, robustness, and scalability. The experimental results show the superiority of our platform.

The rest of this paper is organized as follows. Section II gives a brief introduction to CSI and motivations for this paper. Section III precisely describes the framework of our IoT platform. In Sections IV and V, we present our occupancy detection and recognition systems based on our platform and machine learning techniques. Then the experiments and evaluations are conducted in terms of accuracy, robustness, and scalability, illustrated in Section VI. We compare this paper with a few related advanced human sensing works in Section VII. At last, we conclude this paper and highlight some future trends in Section VIII.

II. PRELIMINARY INVESTIGATIONS ON WIRELESS SIGNALS

At the physical layer of wireless communications, RSSI and CSI are two kinds of useful information for environment perception. As RSSI only exhibits the propagation process by simple signal strength, the RSSI-based occupancy sensing cannot achieve satisfactory results. Though state-of-the-art methods such as PAWS [15] make full use of RSSI, the accuracy of human activity recognition only reaches over 70%. In comparison, CSI is more fine-grained. Previous work [20] builds a convincing model between CSI phase and human speed, which achieves effective result of recognition by training more than 1000 samples. However, in practice, some relatively static activities, such as sitting and lying, should also be considered, and the training phase is too tedious. In this section, we start with a brief introduction to CSI and expand the knowledge of how human activity affects wireless signals by means of preliminary experiments.

A. CSI

In wireless communication, CSI is a representation of the channel properties of a communication link. During the propagation of wireless signals, they are affected by physical environment, which leads to reflections, diffractions, and scattering [21]. These phenomena can be described by CSI. Modern WiFi devices adopt orthogonal frequency division multiplexing (OFDM) at the physical layer and obey IEEE 802.11n/ac standard that permits multiple transmit and receive antennas for multiple-input multiple-output communication. Therefore, CSI reveals fine-grained characteristics of wireless signals combining effect of time delay, amplitude attenuation, and phase shift of multiple paths on each communication subcarrier. Since RSSI is only a superimposition of multipath signals, CSI naturally owns higher resolution. The

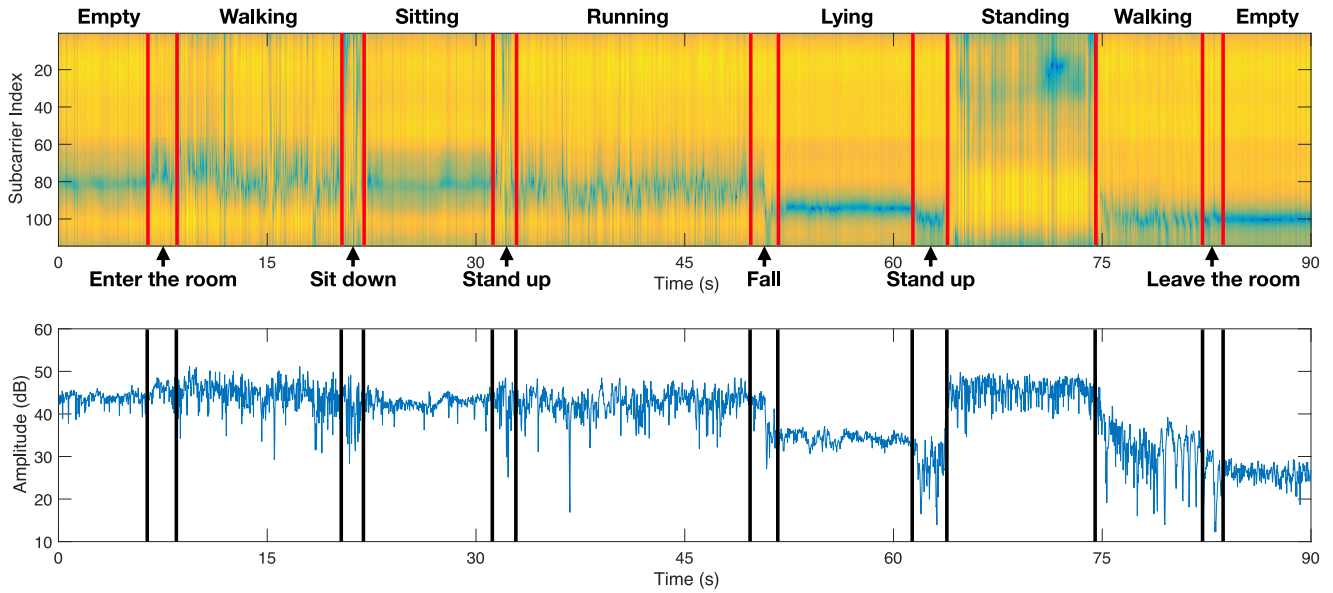


Fig. 1. Visualization of one CSI stream and the 100th subcarrier for a series of different human activities.

WiFi signals can be modeled as channel impulse response $h(\tau)$ in frequency domain

$$h(\tau) = \sum_{l=1}^L \alpha_l e^{j\phi_l} \delta(\tau - \tau_l) \quad (1)$$

where α_l and ϕ_l represent the amplitude and phase of the l th multipath component, respectively, τ_l is the time delay, L indicates the total number of multipath, and $\delta(\tau)$ denotes the Dirac delta function. However, only clusters of multipath components are differentiable because of limited WiFi bandwidth. In this case, the OFDM RX provides a sampled version of the signal spectrum at subcarrier level, which comprises amplitude attenuation and phase shift via complex number. These estimations can be represented by

$$H_i = H_i e^{j\angle H_i} \quad (2)$$

where H_i and $\angle H_i$ are the amplitude and phase of i th subcarrier, respectively.

Theoretically, CSI phase information is supposed to be more robust with less variations. Nevertheless, it is opposite due to hardware imperfections and environmental variations [22]. The carrier frequency of a device can drift by up to 100 kHz for 5 GHz band. In practical applications, it brings about extreme complexity to accomplish calibration and denoising. Thus, we employ amplitude information to design our system.

B. Insights—How Human Activity Affects Wireless Signals

Due to multipath effects, human motion causes the variations of wireless signals that are characterized by CSI amplitude. To link the variation to common human activities, we conduct a preliminary experiment. We deploy two TP-Link N750 routers, each with one antenna, as TX and RX. The TX generates and sends packet every 2 ms and the same frequency of reception in RX. The distance between them is 1.5 m, and the number of subcarriers is 114 in 40 MHz bandwidth with 5G mode. As shown in Fig. 1, a series of

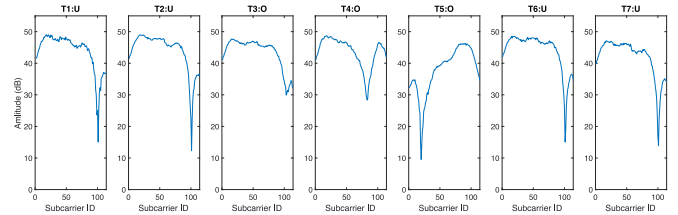


Fig. 2. Curves of CSI subcarrier values from unoccupied (U) to occupied (O) scenarios. We can see that there exists obvious difference of curves between O and U.

activities, including *Walking*, *Sitting*, *Running*, *Lying*, and *Standing*, can be revealed from the CSI stream, in which kinds of activities exhibit distinct statistical characteristics. Moreover, we can even coarsely distinguish different activity from signals of a single noisy subcarrier. However, different part of human body impacts different subcarrier in one TX–RX pair, so we have to make full use of CSI data to recognize the activities more accurately.

However, in Fig. 1, the human presence or not, such as *Empty* and *Sitting*, cannot be distinguished by time-domain variations. Inspired by multipath effect and previous indoor localization work, we investigate the subcarrier difference of static activities shown in Fig. 2. It reveals the frequency information of signal propagation and describes the static activities to some extent. We also find that as long as the layout of the room is fixed, this characteristic is pretty robust. Also, for more considerations, we ever test what degree of activity can result in the changes of wireless signals. The dogs or cats movings are too subtle, which can be regarded as noise.

In summary, the above results indicate that CSI is a fine-grained reflection of the surrounding environment. We can summarize two key insights.

- 1) *Shape of CSI Curve → Human Presence*: The subcarrier curve keeps stable under static circumstances but the occupancy presence affects it to some extent.

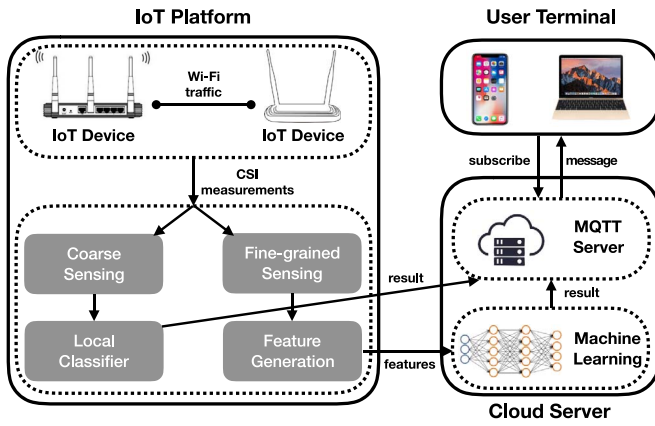


Fig. 3. Overview of the system architecture.

- 2) *Time domain → Human Activity Information*: Time domain analysis of CSI reveals the influence of different activities. Thus, statistical features can be extracted and used for representative learning.

These insights will be utilized for occupancy detection in Section IV and human recognition in Section V, respectively. To enable the applications built on the insights, we first present an agile IoT platform which optimizes the calculation and transmission resources.

III. AGILE WiFi-ENABLED IOT PLATFORM FOR OCCUPANCY SENSING

In this section, we introduce our agile WiFi-enabled IoT platform for occupancy sensing, which consists of an IoT platform for WiFi device, cloud server for complex computations and user terminal for display of results. In the platform design, it is significant to ensure the real time with the consideration of network and computing overload. For example, if CSI measurements are transmitted to the server at a high sampling rate, the successive transmission will put a great deal of burden on the network of the server. Meanwhile, if we process every frame of CSI in the server, it may cause computing resource starvations. After initial experiments and investigations, we design our real-time occupancy sensing system shown in Fig. 3, integrating embedded operating system OpenWrt, lightweight messaging prototype MQ Telemetry Transport (MQTT), and cloud server. The IoT platform refers to two devices supporting WiFi communication, which is installed our innovative OpenWrt firmware to get the capacity of reporting CSI from physical layer directly. To reduce the pressure on network transmission, we conduct some reasonable calculations on the device according to various applications, and these results are transmitted to the server for further processing. The cloud server is responsible for machine learning framework and message distribution as MQTT broker. The sensing results are finally displayed on the user terminal, such as laptops and mobile phones.

A. IoT Platform

The IoT platform takes charge of real-time collection of CSI and primary computations, which enables rapid and

scalable deployment. Conventional CSI tools used in almost all researches of CSI-based sensing, such as Intel 5300 NIC tool [16] or Atheros 9390 tool [23], consist of a commodity router as TX and a mini PC or specific devices with WiFi NIC card as RX. Generally speaking, they are sufficient for research but not for practical applications. There exist two limitations in terms of these platforms. First, the RX has to be laptops with specific network card, which restrains the practicability and scalability. For instance, Wang *et al.* [24] employed more than ten laptops as RXs in the implementation of an indoor localization system. The situation leads to high cost and inconvenience of maintenance, impeding the realization of large-scale systems. The second limitation lies in the resolution of CSI measurements. Intel 5300 NIC tool [16] only provides CSI for 30 out of 56 subcarriers with 20 MHz bandwidth for each TX–RX pair. The CSI measurements are sampled which irreparably loses much necessary information in the physical layer.

To deal with these bottlenecks, we upgrade the Atheros CSI Tool [25] and develop a new firmware for OpenWrt system, which can be deployed in many COTS routers. Our platform only requires two devices with Atheros WiFi module, such as laptop, router, and development board, which boosts the convenience and reduces the cost substantially. The CSI can be directly reported in the OpenWrt system. We choose OpenWrt for it supports a great number of routers and boards. It is so lightweight that much redundancy of normal operating systems is removed. Equipped with the new firmware, the IoT devices have the ability to report fine-grained CSI with complete 114 subcarriers with 40 MHz bandwidth under 5 GHz. Moreover, we can directly conduct some essential computations on the IoT devices.

We aim at building a generic platform where kinds of applications can be developed without involving lots of efforts. Thus, we classify the tasks of occupancy sensing in smart homes into two categories: 1) coarse sensing and 2) fine-grained sensing. To make full use of the computing resources on IoT devices and cloud server, we render our platform in two modes. For coarse sensing tasks including room-level localization and occupancy detection, the computational complexity is tolerable for IoT devices. We manage to implement the algorithms such as local classifier on the board to complete the task. Then the device directly transmits the result to the server and let the users obtain real-time information without using the computing resources of the server. Fine-grained sensing tasks mainly refer to activity recognition and even gesture recognition. These works require advanced signal processing and machine learning techniques to achieve better performance, but it costs enormous computing resources. Furthermore, real-time transmission of CSI measurements can bring about network overload. To tackle these challenges, we propose our recognition approach in Section V, which extracts features from selected CSI data in (7) for transmission. Then the NMF-based classification is conducted in the cloud server. This approach also works as for other state-of-the-art methods, which mostly extract statistical features (such as mean, minimum, and entropy) on the device and then transmit them.

B. Cloud Server and User Terminal

With the rapid development of cloud computing, the cloud server has huge advantages in workload resiliency and stability. In our system, we employ the cloud server, which presides over machine learning computations as well as message forwarding. In order to optimize network communication between lightweight OpenWrt and the server, MQTT is introduced in our system. MQTT is an extremely simple lightweight messaging protocol, which is designed specially for constrained devices and low-bandwidth or high-latency networks. It has been successfully deployed in satellite links, medical, and miniaturized devices [26]. With the help of MQTT, we manage to overcome the network overload.

In MQTT, the message is transmitted via TCP/IP protocol and it provides bidirectional connections in order without prejudice. The MQTT is integrated in our system as follows.

- 1) There are three roles in MQTT, including *publish*, *broker*, and *subscribe*. The cloud server, IoT platform, and user terminal play the role of *broker*, *publish*, and *subscribe*, respectively. The cloud server (*broker*) forwards the specific message to the user (*subscribe*) when IoT platform (*publish*) pushes information to the cloud server. Our IoT platform (*publish*) generates sensing results and transmit the results to cloud server. Then the users (*subscribe*) are able to obtain required message. In our test, MQTT ensures the real-time notification in terms of IOS, Android and Web terminals.
- 2) For MQTT *broker*, it offers quality of service (QoS) regarding different clients. In our system, we enable the highest QoS for transmission.
- 3) MQTT messages are delivered via matching *topic name*, which is a tag in each message. The IoT and subscribed devices of same user share same *topic name*. By this way, we make it possible to deploy our system in large scale of users.

After the MQTT pushes the message to the user terminal, the user interface should display real-time sensing results and be compatible with multisystems. In light of this, we develop the user interface based on Web, which supports most smartphones and laptops.

Based on our innovative platform, we propose two novel approaches to realize real-time applications. The occupancy detection is regarded as coarse sensing case, and it is purely implemented on the IoT device. In comparison, human activity recognition is a fine-grained occupancy sensing technology, and its machine learning framework is running on the server.

IV. COARSE SENSING—OCCUPANCY DETECTION

Precise detection of occupancy is indispensable to improve the efficiency of building management system [27]. It can be utilized for intrusion detection in commercial area, thermal comfort adjustment in smart building, and elderly monitoring at home. Traditional methods mostly rely on light, temperature, and CO₂ measurements [3]. Based on our IoT platform, we propose an occupancy detection application as a case study of coarse sensing.

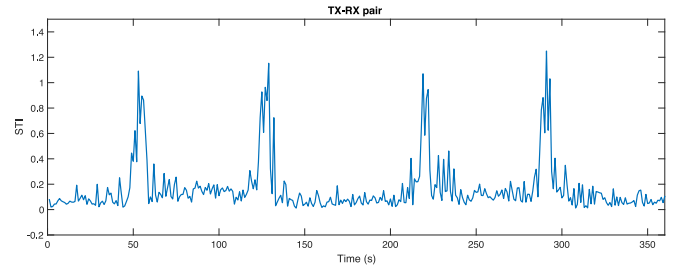


Fig. 4. STI values of one TX-RX pair. The occupants appear for four times indicated by four peaks.

As mentioned in Section II, we can know the situation of human presence by CSI subcarrier curves in Fig. 2. Through further validation, the curve without any denoising is stable, so we only need to explore a method to compare the shape similarity between adjacent CSI curves in time series. Hence, we employ signal tendency index (STI) [28] to compare the shape similarity of CSI curves. It owns the invariance of translation and scale so that it can compare the tendency and waveform from the statistical point of view. Suppose $H_1^t, H_2^t, \dots, H_n^t$ is a CSI vector at timestamp t . STI superimposes these curves by optimally translating and uniformly scaling. The translation step produces

$$H_1^t - \bar{H}^t, H_2^t - \bar{H}^t, \dots, H_n^t - \bar{H}^t \quad (3)$$

where $\bar{H}^t = (1/n) \sum_{i=1}^n H_i^t$. Then comes the scaling step as follows:

$$\hat{H}^t = [H_1^t - \bar{H}^t, H_2^t - \bar{H}^t, \dots, H_n^t - \bar{H}^t] / \sigma(H^t) \quad (4)$$

where $\sigma(H^t) = \sqrt{\frac{1}{n} \sum_{i=1}^n (H_i^t - \bar{H}^t)^2}$. The transformed object $\hat{H}^t$ can be used for similarity comparison. As we can derive $\widehat{H^{t-1}}$ in the same manner, it is simple to compute STI to compare two shapes of vectors

$$S^t = \|\hat{H}^t - \widehat{H^{t-1}}\| \quad (5)$$

where $\|\cdot\|$ denotes the Euclidean norm. Small S suggests that the two curves are similar while large one indicates that they are uncorrelated. As shown in Fig. 4, the STI values with human presence are much higher than those under the unoccupied situations. We can infer that the shape similarity of CSI curves indeed makes a distinction between occupied and unoccupied situations. An empirical threshold $\text{th}_d = 0.8$ selected through cross-validation is introduced to classify the situations. The whole algorithm of occupancy detection is described in Algorithm 1. We implement the prototype on the IoT device and demonstrate that each time of similarity computing can be completed at a frequency of 10 Hz, which is sufficient for occupancy detection. The detailed evaluation in Section VI validates its robustness and effectiveness.

V. FINE-GRAINED SENSING—HUMAN ACTIVITY RECOGNITION

Human activity recognition is a crucial technology which enables wide applications in smart homes such as health surveillance and behavior analysis. As we know, the human

Algorithm 1 Occupancy Detection Using STI

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1: while IoT device is running well do
2:   Obtain CSI measurements  $H^t$  at timestamp  $t$ 
3:   Calculate the STI value  $S^t = \|\hat{H}^t - \hat{H}^{t-1}\|$ 
4:   if  $S^t > th_d$  then
5:     Send result of human presence to MQTT server
6:   else
7:     Continue
8:   end if
9: end while

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activity is captured by wireless signals and characterized by time-domain features. On account of this, we build a real-time activity recognition system based on our IoT platform.

Existing systems [29], [30] achieve remarkable results in ideal indoor environments by means of extracting statistical features and training large number of samples. However, there still remain two limitations. First, the basic activity such as *Walking* can occur at any positions around the routers. It is unknown whether the statistical features display similar characteristics. Second, in practice, the collection of tremendous number of training samples costs too much labor and time. To promote this technology for practical applications, these challenging problems have to be overcome. In this section, we present our user-friendly location-independent human activity recognition approach as a fine-grained sensing case in our IoT platform. It can accurately recognize various human activities at different positions according to customized demands.

A. Activity Detection

To capture the time-domain information, we use a sliding window with time length L to get samples from raw CSI streams. After we obtain a CSI matrix $M_{N_s \times L}$ from IoT devices, where N_s is the number of subcarriers, it is initially examined to justify whether an activity is happening during this period. If the human motion is detected in this window, then we send it to our server for further processing.

For every CSI matrix M , it includes N_s bins along the row order, and each bin contains L CSI measurements. As human motions always lead to variation of subcarrier value of CSI, we can infer whether an activity is happening by its fluctuation degree. The variance ν_k of each bin is calculated and we obtain a cumulative moving variance as $\Phi = \sigma_{k=1}^{N_s} \nu_k$. Then an empirical threshold σ_v is introduced. When $\Phi > \sigma_v$, it indicates the occurrence of a human activity. It could have been improved by an adaptive threshold [31]. However, we intend to make it simpler and more effective with lower power consumption. We may use adaptive threshold in the future work as the computing ability of IoT device is boosted.

B. Noise Removal

Once an activity is perceived, it should be transmitted for classification. However, after observation, we find that raw CSI data has intrinsic noise and thus need preprocessing for high level recognition. We eliminate the inherent noise in time domain with a setup of interpolation.

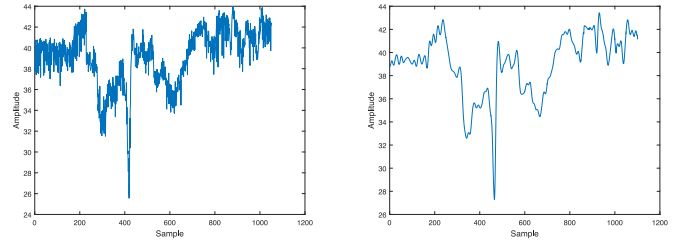


Fig. 5. Comparison of raw and denoising data on subcarrier #20 in CSI stream.

1) *Interpolation*: During the wireless communication between TX and RX, the RX may report empty CSI measurements due to packet loss. To remedy the loss, we utilize linear interpolation to guarantee the quality of measurement at each timestamp and stationary interval of successive CSI frames.

2) *CSI Smoothing*: As the transmission power varies during time, some high frequency noise such as Gaussian noise is caused in the time domain. To deal with it, we utilize discrete wavelet transform (DWT) to denoise it. Wavelet transform uses various transforms $\psi_{a,b}(t) = a\psi(t-b)$ obtained from a basic function through scaling and translation in terms of parameter a and b , respectively. The wavelets coefficients of our CSI time sequence $r(t)$ can be calculated by

$$W(a, b) = \int_{-\infty}^{\infty} \psi_{a,b}^*(t) r(t) dt \quad (6)$$

where $*$ is the complex conjugate and *Symlet* is chosen to be the basic function ψ . After DWT-based denoising, we filter the coefficients and calculate the sanitized CSI data $\hat{M}$ by the inverse transformation. The comparison before and after the preprocessing phase is shown in Fig. 5. By means of noise removal, the sanitized CSI data is ready for activity recognition.

C. CSVD-NMF Scheme for Activity Recognition

1) *Activity Extraction via Class Estimated Basis Space Singular Value Decomposition*: We aim to extract activity from CSI data wherever it happens. The information of human activity mainly refers to CSI caused by multipath effect. In previous work [32], it is demonstrated that the CSI dominant component of CSI lies in that of line-of-sight (LOS) path, which does not reflect much human information. Thus, it is sensible to extract useful component by eliminating LOS part. However, singular value decomposition (SVD) cannot be directly used because the base space of the singular vectors corresponding to CSI of different activity is inconsistent. In this case, we propose a CSVD-based reconstruction method to eliminate main components of static paths in CSI data.

For each CSI matrix M in $\mathbb{R}^{S \times L}$ space, we can compute the SVD of it, $U\Sigma V^T = M$. Note that $U^T U$ and $V^T V$ both yield the identity matrix so they both have orthonormal columns. The matrix Σ is a diagonal matrix whose diagonal elements $\{\sigma_1, \sigma_2, \dots, \sigma_\gamma\}$ are singular values in decreasing order, $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_\gamma \geq 0$. γ is the number of singular values. The prominent property of the SVD is that when

we reconstruct an approximation to a matrix Q , the singular values perform decreasing importance. Explicitly, we define rank- K approximation to have first K singular values in Σ but others are zero. It can be proven that rank- K approximation is always the closest rank- K matrix to Q . That is $Q_K = \operatorname{argmin}_A \|Q - A\|_F$, where the operation $\|\cdot\|_F$ is the Frobenius norm.

In our application, to eliminate the main component, we specify the first singular value to be zero and reconstruct an approximation as follows:

$$M_{\sigma_1=0} = \sum_{i=2}^{\gamma} \sigma_i u_i v_i^T \quad (7)$$

where u_i and v_i are the singular vectors corresponding to the singular value σ_i . We can find the similarity of $M_{\sigma_1=0}$ of the same activity in different locations. Obviously, without the dominant component, the remained CSI preserves more characteristics of human activity. In the training phase, we calculate the average CSI of each class and reconstruct the CSI for NMF classification.

2) *Automatic Feature Extraction and Classification Using Non-Negative Matrix Factorization*: We employ NMF to conduct automatic feature extraction and classification. It can be observed that most discriminative features of the CSI are determined by part of pixels in the whole frame. Although hand-crafted features can be mined in time and frequency domain, the representative learning by hands may lose unobservable characteristics. Therefore, we develop this method to learn a robust representation for classification automatically. Surprisingly, as NMF-based classification does not requires large scale of training samples, it permits fast extension to new types of activities with only several times of calibration.

a) *Training phase*: NMF employs non-negative constraint to learn part-based representation. After normalization to guarantee non-negative property, we can decompose each class of training data obtained by CSVD, $V_{S \times L} = [v_1, v_2, \dots, v_L]$, to its subspaces, i.e., $V \approx WH_c$, where $\sum_{i=1}^n v_{ij} = 1, j = 1, 2, \dots, L; v_{ij} \geq 0, W \in \mathbb{R}^{S \times K}$ is the base representation of CSI and $H_c \in \mathbb{R}^{K \times L}$ denotes the coefficient. K is the number of basis of feature subspace W . Each column $v_i \in \mathbb{R}^{S \times L}$ represents a training sample for one activity. This problem can be solved using divergence approximation $Y = WH_c$. Then it is equivalent to the optimization problem

$$\begin{aligned} \min_{W, H_c} D(V \| WH_c) &= \sum_{i,j} \left(v_{ij} \log_n \frac{v_{ij}}{y_{ij}} - v_{ij} + y_{ij} \right) \\ \text{s.t.} \quad \sum_{i=1}^n w_{ij} &= 1 \quad \forall j \text{ and } W, H_c \geq 0. \end{aligned} \quad (8)$$

The optimal result W, H_c are regarded as feature subspaces matrices for one activity and we train each one in the same manner. Note that the number of training samples can be very small and the extension of new class does not need training classifier for the whole dataset. When the system is deployed in the new environment, it only needs a few samples to retrain the classifier, which is also convenient and effective to use.

Algorithm 2 Training Process of CSVD-NMF

- 1: Select a target action to be measured
 - 2: Execute the action and obtain a CSI measurement
 - 3: Conduct SVD for denoised CSI matrix and eliminate the dominant component
 - 4: Reconstruct an approximation CSI matrix
 - 5: Yield the subspaces matrix by reconstructed CSI matrix
 - 6: Store the subspaces matrix and return to Step 1
-

b) *Classification phase*: For a new test sample X , we project it to the feature subspace of k th activity feature subspace by $H_x = W_k^{-1}X$ and H_x represents the reconstruction of X using the basis of the new subspace W_k . Then we compare the coefficients H_x with the original H_k , and define the normalized error as the classification criteria

$$\varepsilon_k = \frac{\|H_x - H_k\|}{\|H_x\|}, \quad k = 1, 2, \dots, c \quad (9)$$

where c is the total number of training samples. This measures the similarity between the current CSI measurement and each activity sample, which is considered as the distance for K -nearest neighbors (KNNs) algorithm. Finally, X belongs to the most probable class in terms of its nearest neighbors. Thus, the online classification only needs matrix multiplication rather than costly hand-crafted feature extraction. The training process of CSVD-NMF-based recognition is concluded in Algorithm 2.

VI. IMPLEMENTATION AND PERFORMANCE EVALUATION

A. Prototype System

We implement a prototype of our IoT platform based on two TP-LINK N750 wireless dual band routers as IoT devices, and a cloud server as computing center. Two routers, used as TX and RX, are upgraded to our innovative OpenWrt system so that CSI is reported on board directly. Note that previous work [33], [34] also employ this mode of access point. Nevertheless, this previous work leverages normal OpenWrt firmware without any modifications in the physical layer. In our architecture, TX with one antenna was set to 802.11n AP mode at 5 GHz with a 40 MHz bandwidth and the client, RX, with three antennas, was connected to this WiFi network. All the routers are deployed at a height of 1.5 m. In this case, multiple clients are allowed in our system, which can form several pairs of network and increase the coverage. Hence, the number of CSI streams is $1 \times 3 \times 114$ and we prototype our occupancy detection and human activity recognition system based on this platform. We conducted the evaluation in an office zone (5.6 m $\times$ 9 m) and a conference room (5 m $\times$ 7 m).

B. Evaluation of Occupancy Detection

For coarse sensing task with the low complexity of Algorithm 1, we implement the occupancy detection function directly on IoT devices. It can detect the presence of any number of occupants. The source codes are written in C languages

TABLE I
OCCUPANCY DETECTION PERFORMANCE

Antenna	Frequency (Hz)	ODA (%)	ODR (%)	UDR(%)
RX1	30	93.2	95.2	92.4
RX2	30	91.5	93.1	89.6
RX3	30	94.8	96.3	93.8
RX 1,2,3	22	96.8	98.2	95.1

and they are cross-compiled for the OpenWrt system. We evaluate CSI-enabled occupancy detection in terms of accuracy, computing speed, and antenna usage. For the evaluation of accuracy, we use three metrics as follows:

1) *Overall Detection Accuracy (ODA)*:

$$ODA = \frac{\text{corrected detected samples}}{\text{all samples}}.$$

2) *Occupied Detection Ratio (ODR)*:

$$ODR = \frac{\text{detected occupied samples}}{\text{true occupied samples}}.$$

3) *Unoccupied Detection Ratio (UDR)*:

$$UDR = \frac{\text{detected unoccupied samples}}{\text{true unoccupied samples}}.$$

As shown in Table I, we evaluate the accuracy using three single antenna as well as all antennas. It is obvious that when we use CSI of only one antenna, the computing time is low but the accuracy only reaches 91%–94%. Then with CSI data of three antennas (3×114 subcarriers), the accuracy runs up to 96.8% and the computing frequency is about 22 Hz which is fairly acceptable. We also discover that the ODR is a little higher than UDR under all circumstances. It is because our method is more sensitive to human motion in LOS and has limited range. When human motions occur around the range border, it is hard to distinguish the presence. This results in some wrong detection of unoccupied situation. In summary, based on all subcarriers of three antennas and our IoT devices, our CSI-enabled occupancy detection achieves high accuracy with satisfactory frequency. The notification of presence is then forwarded to the user end by our cloud server.

C. Evaluation of Human Activity Recognition

During our activity recognition experiments in both office zone and conference room, one user was asked to perform different activities near one TX–RX pair of routers, while other people coexisted in the same room. These people were sitting and working in their cubicles and relatively away from the TX–RX pair and their interference can be considered as background noise. We configured the experimental setup in this way because the current objective of our system is to provide personal device-free human activity recognition for realizing tailor-made services in the smart home and smart buildings. To this end, we evaluate the accuracy, extensibility, and robustness in an indoor environment. The empirical parameters are set as: $N_s = 3 \times 114$, $L = 1000$, $K = 5$ in NMF and four nearest neighbors in KNN. We adopt three metrics to evaluate the recognition accuracy.

TABLE II
COMPARISON OF TPR OF DIFFERENT APPROACHES

Methods	Sitting (%)	Walking (%)	Running (%)	Overall (%)
E-eyes	94.13	88.91	85.06	89.37
CARM	97.58	92.82	91.74	94.05
Ours	96.21	99.52	96.33	97.35

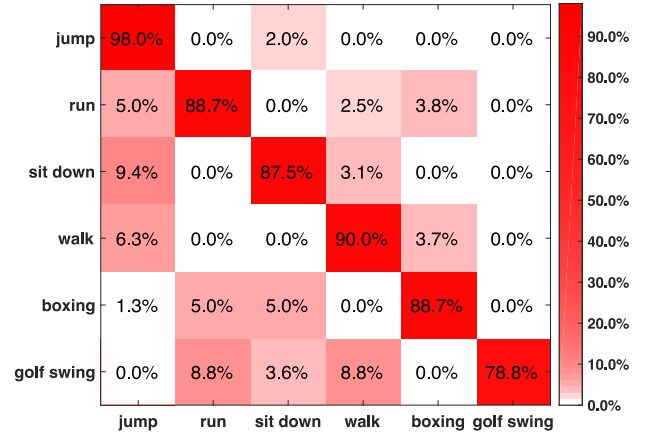


Fig. 6. Confusion matrix of recognition accuracy.

- 1) *True Positive Rate (TPR)*: The ratio of the number of times for correctly recognizing an activity to the total number of activities.
- 2) *False Positive Rate (FPR)*: The fraction of cases that are incorrectly recognized as an activity among all testing samples other than activity.
- 3) *Confusion Matrix*: Each cell represents the recognition accuracy in terms of one activity.

1) *Recognition Accuracy*: The data of activities, including *Jump*, *Run*, *Sit Down*, *Walk*, *Box*, was collected in eight fixed locations around the RX one to two meters apart, and each class has 20 samples in each location (totally 160 samples). We randomly use 20 samples in five positions for training while other 140 samples in eight positions for testing. After cross validation, the result is performed by confusion matrix. To compare its performance with two state-of-the-art CSI-based activity recognition methods, E-eyes [30] and CARM [29] in terms of *Sitting*, *Walking*, and *Running*, we keep the same setting as these methods. We use five sample in all positions for training and others as testing. As shown in Fig. 6, the accuracy of *Jump* reaches 98% and the other four activities achieve about 90% accuracy. The total average accuracy is calculated as 90.6% with only 20 training samples. In Table II, we can see that our method improves the overall TPR by 3.3% and 7.98% over E-eyes and CARM, respectively. The reason is that they adopt limited hand-crafted features of coarse CSI data with 30 subcarriers information.

2) *Extension Evaluation to New Activities*: To demonstrate the performance of extension ability, we add a new challenging activity *Golf swing*. The average accuracy still reaches more than 78% in terms of only five training samples shown in the last row in Fig. 6, while the advanced methods such as [35] utilize 150 samples of each activity to train the model. We also investigate the relationship between number of samples and

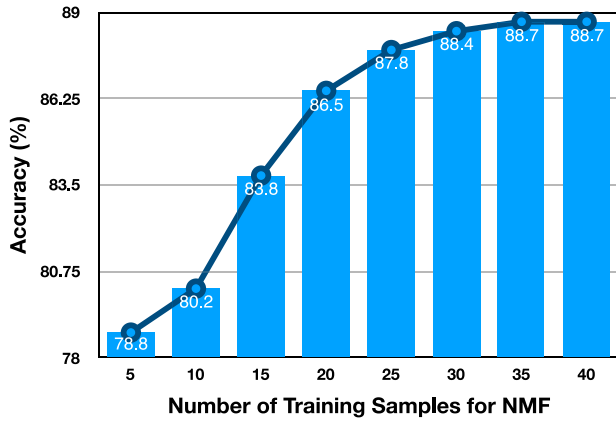


Fig. 7. Relationship between number of training samples and recognition accuracy in terms of *Golf Swing*.

TABLE III
COMPARISON OF FPR OF DIFFERENT APPROACHES

Methods	Sitting (%)	Walking (%)	Running (%)	Overall (%)
E-eyes	5.53	10.93	12.33	9.60
CARM	2.24	6.48	8.22	5.65
Ours	2.35	0.84	3.63	2.73

accuracy in Fig. 7. The accuracy is improved as the number of training samples increases, but over 30, the improvement is not obvious and it may bring about overfitting.

3) *Robustness*: Table III demonstrates the comparison of FPR between three activities. Our method achieves best FPR of 2.73% over E-eyes of 9.60% and CARM of 5.65%. The robustness of our system lies in both the novelty of automatic selected features and the KDE-based detection method that distinguishes activity from the background. Adopting traditional detection method based on variance, sometimes the system detects false motion caused by environmental changes rather than human beings.

D. Multiple Users Evaluation

To further show the use case of our IoT system, we conduct an evaluation by combining two methods using several TX–RX pairs. The objective is to analyze the attendance and recognize the activity in the multiuser mode. The system was deployed in a 5.6 m × 4.5 m office zone and there were totally three TX–RX pairs. Each pair is able to detect the state of the user in his cubicle. Shown in Fig. 8, the top two pairs were in the cubicle 1 (left) and 2 (right) with a standoff distance of 1 m, while the bottom one was in the cubicle 3 with a 2 m distance. We collected data of three activities, including *Sitting*, *Walking*, and *Running* and the *Empty* or not was reported by occupancy detection approach. During the data collection, three users were performing their activities in their cubicle zones simultaneously. We obtain 50 samples of each activity for each user, and employ our method to do the classification with same empirical parameters as former evaluations.

As depicted in Fig. 9, we get an overall accuracy of 96.4% for the occupancy detection and 96.3% for three activities. From the perspective of activity, similar to single-user evaluation, *Walking* gets the best accuracy of 98.0%, and then 95.6%

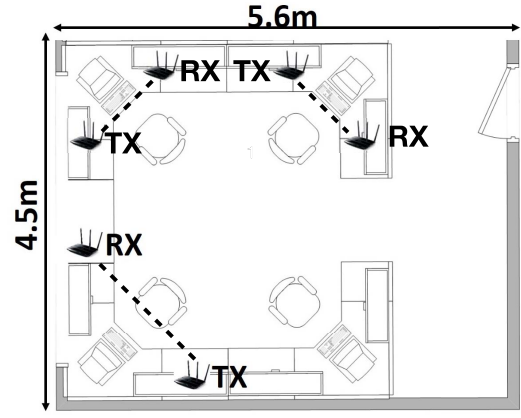


Fig. 8. Layout of multiple users evaluation.

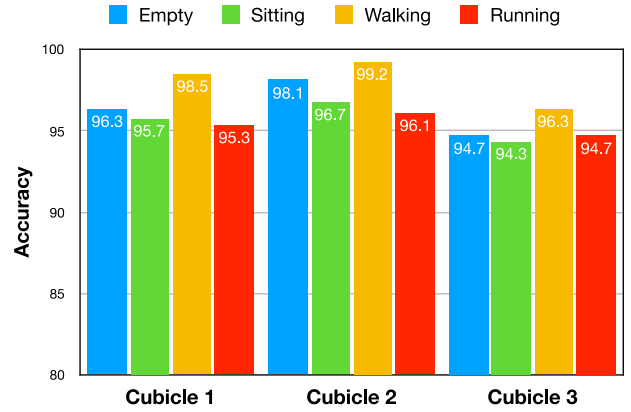


Fig. 9. Performance of multiple users evaluation.

and 95.4% for *Sitting* and *Running*, respectively. Compared to the result in Table II, it drops by 1%–2% around due to interference and environmental noise. For these reasons, it is easy to discover that the accuracy of cubicle 3 is the worst. It is because that the distance between TX and RX is positive relevant to its sensing range. In cubicle 3, its pair has the largest range, and thus it is easily affected by the surrounding changes by others. Similarly, other two pairs have less sensing range and their CSI data can be mostly dominated by their users. The multiple user evaluation demonstrates that the multiuser situation can be handled by multipairs. Each pair of TX–RX not only captures the activity but also reflects the location of the happening activities, which will bring greater significance for smart homes in reality.

VII. RELATED WORK

In the recent few decades, occupancy sensing, especially human activity recognition, has aroused a great deal of attention in many research fields and its major objective is to sense and recognize occupant activities by extracting features from environmental sensors.

Many approaches are proposed and they can be classified by their devices including wearable sensors, computer vision, and WiFi. Each approach has advantages and disadvantages. Wearable sensors such as accelerometer sensor are

TABLE IV
COMPARISONS OF WiFi-BASED HUMAN SENSING SYSTEMS IN SEVERAL METRICS

System	Objective of sensing	Methodology	Performance	Complexity	Scalability	Devices
PAWS [8]	Empty, walking, sitting, standing, fallen, sleeping, running	RSSI-based, fusion algorithm with two features	72.47%	Training needed	Easy	COTS WiFi routers
WiSee [30]	Gestures: push, drag, kick, circle, punch, bowling, strike, dodge	Doppler shift, pattern matching	94%	No training	Medium	USRP-N210 hardware
WiGest [31]	Gestures: open/close, swipe left/right, wipe, hold over, towards	RSSI-based, Segmentation and Matching	87.5%	No training	Easy	Cisco Linksys X2000 and Netgear N300
E-eyes [20]	Daily Activity: cooking, washing dishes, taking a bath, sleeping, etc	CSI-based, classification of in-place and walking activity	96% with multiple links	Clustering needed	Difficult	Laptops with Intel 5300 NIC card
CARM [21]	Empty, walking, sitting, standing, running	CSI-based, CSI speed model and PCA based features	96% with multiple links	Training needed with PCA	Difficult	Laptop with Intel 5300 and TP-link TL-WDR7500
WiFall [32]	Activity: Falling	CSI-based, one-class SVM with 7 features	87% detection accuracy with 18% false alarm	Training needed	Difficult	Intel WiFi Link 5300 802.11n NIC and require modifications
WiWho [35]	Gait of single person	CSI-based, decision tree based classifier with time and frequency domain features	80% with 6 people	Training needed	Difficult	Laptop with Intel 5300 NIC and Asus RT-AC66U routers
Our work	Activities: Jump, run, sit down, walk, box, golf	Fine-grained CSI-based, CSVD-NMF classification with KNN	90.6% with 20 training samples	Training needed	Easy	TP-Link N7500 and more COTS WiFi routers

commonly used in mobile devices. Due to its high frequency and convenience, Chen *et al.* [12] developed an effective algorithm to recognize human activities based on principal component analysis (PCA) and support vector machine. However, people may not carry mobile devices all the time. With the rapid development of computer vision, vision-based methods emerge and achieve high accuracy using deep learning and convolutional networks [7], [36]. Though it achieves high accuracy of recognition, it can only detect a pretty limited range and it is easily affected by illumination and occlusion. In smart building, kinds of sensors are collected for human sensing including thermal sensors, CO₂ sensors, etc [37]. Nevertheless, this kind of system requires large installing scale of sensors and thus brings about high cost.

To attain a better tradeoff between accuracy and cost, WiFi-based methods come into existence. They employ ubiquitous WiFi devices for activity recognition with high scalability and practicability. We present a brief summary and comparisons of several state-of-the-art methods in Table IV. We compare both RSSI-based and CSI-based methods in terms of activity, methodology, complexity, performance, scalability, and practicability. RSSI-based methods, such as PAWS [15], WiSee [38], and WiGest [39], are effective ways to detect human activity but the accuracy is not satisfactory due to the restriction of resolution of RSSI. CSI is more fine-grained that has been utilized for various applications, such as occupancy detection [40], activity recognition [29], [30], [41], [42] (E-eyes, CARM, DeepSense, and FALAR), sedentary behavior monitoring [43], falling detection [44] (WiFall), crowd counting [45], indoor localization [24], [46], [47], and person identification [48]–[50]. These methods realize fine-grained recognition of human activity, but they are deficient in availability due to the lack of scalable platform. Our innovative CSI-enable IoT platform fills the gap and provide a chance to promote these applications for practical usage. For conventional WiFi-based activity recognition system, they employ either PCA or statistical features to classify different

activity, while we address the problem by an automatic feature selection approach followed by an effective classification phase. This paper achieves similar results with only several training samples, which is more reasonable considering its real case and deployment.

Although our platform makes the CSI-based occupancy sensing take a stride forward, it is still limited by inherent characteristics of WiFi signals. First, the sensing range is restricted by the coverage of WiFi signals. Second, for the activity recognition, multiuser cases require more TX–RX pairs. For instance, the current settings in our multiuser experiment employ many routers, which are not cost-effective. Whether one pair can detect multiple users will be one of our major work in the future. Anyway, this technology can be integrated with other systems such as camera-based methods, which can complement each other. We believe that a fusion solution could be convincing in achieving occupancy sensing with reliable performance.

Except these limitations, some concerns should also be noticed as follows.

- 1) First, we might be worried about the health problem addressed by wireless signals. On the one hand, the health problem is a side effect indeed, as well as mobile phones and PCs. On the other hand, we leverage existing WiFi router deployed in smart homes, which can not only provide Internet connection but also sense the humans. The commercial routers accord with the international safety standard. If we install other devices such as RF or radar, the radiation will be more severe. Considering the existing WiFi infrastructures in homes, we believe that this technology is rather acceptable in reality.
- 2) Second, the sensing data may occupy the network bandwidth. This can hamper the availability of our technology. Actually, the network traffic will suffer from our transmission and there exists a bandwidth loss. However, this is relevant to our package transmission

rate. When we only extract CSI with 40 Hz such as our method, we have not found any side effects of network traffic. Specifically, for 802.11n, the practical transmission speed is 100 Mb/s while our extraction of CSI with 342 subcarriers occupies only about 0.1 Mb/s. However, existing approaches improve the performance by increasing the transmission frequency, and consequently the traffic becomes worse. For instance, previous methods [29], [30] employs data with at least 2000 Hz, which occupies at least 5 Mb/s and is not so acceptable using existing WiFi infrastructures.

VIII. CONCLUSION

In this paper, we design a CSI-enabled IoT platform for occupancy sensing in smart homes, and present two innovative applications based on it. Our platform only employs commodity WiFi routers and thus can be easily deployed in smart homes. It consists of three parts, referring to IoT platform, cloud server, and user end. We first integrate OpenWrt system with CSI tools and then the CSI reported in the IoT devices will be transmitted to cloud server for processing. Finally, the results are displayed in the user end. To demonstrate the availability, we implement two cases including occupancy detection and human activity recognition. We present the coarse case, occupancy detection, by means of exploiting the CSI curve features in terms of the presence of persons. It is completely realized in the IoT device and the evaluation shows an accuracy of 96.8% with 22 Hz. For more fine-grained case, we present a CSVD-NMF-based human activity recognition method, which aims to improve the extendability and robustness. The evaluation demonstrates that our method achieves more than 90% accuracy with only 20 training samples for each class. As a result, our platforms provide a significant opportunity to enable many CSI-based applications.

For the future work, we mainly consider multiuser situations and data fusion approach. It is possible to employ very costly access points and network cards, mini-controllers such as ESP8266. This can provide more connections and more detailed sensing data. As various sensing technologies can compensate for each other, it is believed that fine-grained sensing can be achieved by fusion of these approaches.

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